HOW ARE GENETIC DISORDERS PASSED ON?

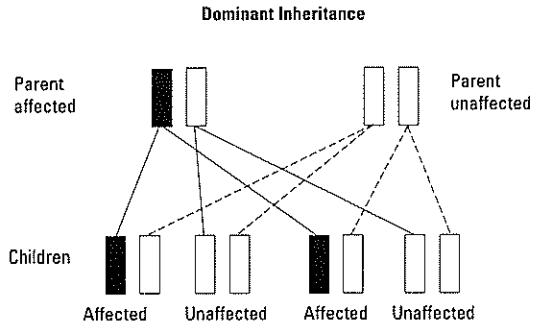
Genetic disorders caused by defects in single genes show three common patterns of inheritance.

A: DOMINANT PATTERN OF INHERITANCE

If a dominant gene is faulty, it will usually show an effect even if there is a working copy of the gene on the other paired chromosome.

Huntington's Disease is an example of a disorder caused by a faulty dominant gene. A person who inherits one faulty copy of the gene which causes Huntington's Disease will almost certainly develop the disorder.

In the diagram below, you can see how each and every child with one affected parent has a 50% (1 in 2) chance of inheriting a faulty copy of the gene.



KEY
- Affected indicates presence or development of the disorder
- Normal Chromosome
- Chromosome bearing a faulty copy of the gene

B: RECESSIVE PATTERN OF INHERITANCE

If a recessive gene is faulty, it will usually show little or no effect unless the same recessive gene is faulty in **both** of a pair of chromosomes.

Cystic Fibrosis is an example of a disorder caused by a faulty recessive gene. A person who inherits *two* faulty copies of the gene (one from each parent) will have Cystic Fibrosis. But a person who only inherits *one* faulty copy of the gene will be a carrier. He/she will not have Cystic Fibrosis but can pass a copy of the faulty gene onto any children he/she may have.

Diagram A illustrates what can happen if **one parent is a carrier** of a faulty copy of the recessive gene which causes Cystic Fibrosis. Each and every child has a 50% (1 in 2) chance of inheriting *one* faulty copy of the gene and being a carrier.

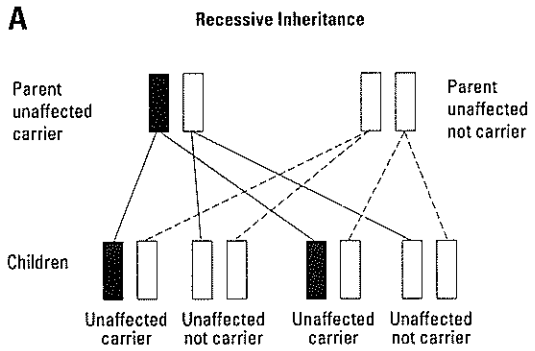
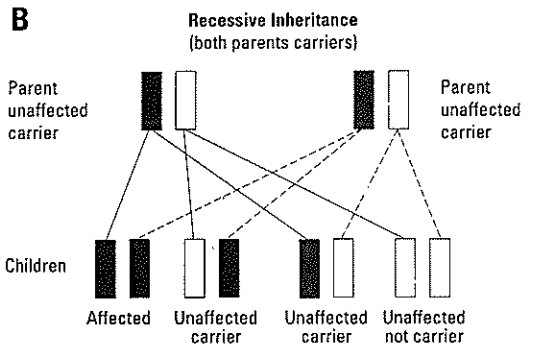


Diagram B illustrates what can happen if **both parents are carriers.** Each and every child has:

- a 25% (1 in 4) chance of inheriting *two* faulty copies of the gene (one from each parent) and having Cystic Fibrosis
- a 50% (1 in 2) chance of inheriting *one* faulty copy of the gene and being a carrier.



C: X-LINKED PATTERN OF INHERITANCE

X-linked genetic disorders are caused by a faulty gene on the X (or female sex) chromosome. Haemophilia is an example of an X-linked genetic disorder.

Females who carry a faulty copy of the gene which causes haemophilia are usually unaffected because they have a working copy of the gene on their other X chromosome.

But males inherit an X chromosome from their mother and a Y chromosome from their father. So if a male inherits a faulty copy of the gene on his X chromosome, he has no working copy of the gene on his Y chromosome to override the faulty copy and will have Haemophilia.

A man with Haemophilia never passes the faulty gene onto his sons because they inherit a copy of his Y chromosome. But all his daughters will be carriers because they inherit a copy of his X chromosome containing the faulty gene.

If a carrier mother decides to have children, all her daughters have a 50% (1 in 2) chance of inheriting a copy of the faulty gene from her and of being a carrier. All her sons have a 50% (1 in 2) chance of inheriting a copy of the faulty gene from her and of having Haemophilia.

The diagram below illustrates an example of X-linked inheritance when the father is unaffected and the mother is a carrier.

